

Assignment 2: Knowledge Representation and Reasoning (KRR)

Due date: Sunday, 20 April 2025 (23:55).

Evaluation: 100 marks = 17%.

Submission: Moodle.

Submission instructions and late submission policy: In Moodle.

For this assignment, *there are no restrictions on the use of generative AI technologies*. However, whenever used, generative AI must be acknowledged. For further information, please, refer to the current university policy on the use of generative AI.

1 Propositional Logic

[15+15 = 30]

Consider the following propositional logic sentences:

- $\phi_1 = A \rightarrow ((A \rightarrow B) \vee (B \rightarrow C))$
- $\phi_2 = (A \wedge B) \leftrightarrow ((B \rightarrow C) \rightarrow ((C \wedge A) \vee A))$
- $\phi_3 = (A \leftrightarrow B) \wedge (\neg(A \wedge \neg C) \vee (B \leftrightarrow D))$

Relative to these sentences,

- (a) Show clearly whether they are either valid, satisfiable (but not valid), or unsatisfiable; in each case, justify your answer. [15]
- (b) Convert them into Conjunctive Normal Form (CNF) and indicate if they are the Horn Clause. [15]

2 Backward/Forward Reasoning

[10+10+10 = 30]

Convert the expressions below into propositional logic sentences: [10]

1. If a person is likely to vomit and looks pale and is thirsty, then is sick.
2. Always, a person does not have a high temperature or has slept well or is thirsty.
3. If a person does not look pale, then feels well or has slept well.
4. If a person has a fever, then their temperature is high.
5. It is known that when a person looks pale, then they should drink water.
6. If a person has a high temperature and does not feel well, then is likely to vomit.
7. It is always the case that a person does not feel well or does not have a fever.
8. If a person has slept well, then is not exhausted.

With the propositional logic sentences obtained from the above expressions in English, show whether or not it can be proved that a person is sick if one assumes both that the person has a fever and that the person is exhausted. Do this reasoning using:

- (a) Forward chaining; and show the graph resulting from the application of the rules. Provide a table showing which variables are in your AGENDA, are INFERRED, and their COUNT throughout the application of the algorithm. [10]
- (b) Backward chaining; and provide a sequence of rules to prove the goal. [10]

- S: is Sick
- V: likely to Vomit
- P: looks Pale
- T: is Thirsty
- H: has High temperature
- NS: has Not Slept well
- NF: does Not Feel well
- F: has Fever
- W: drink Water
- E: is Exhausted

3 First-Order Logic

[20+10+10 = 40]

Use resolution to check whether the goal “G” below can be proved or not. To do so, show:

- (a) how to construct all necessary clauses, [20]
- (b) the application (all the steps) of the proof method, [10]
- (c) which most general unifiers are used (in case a proof can be found). [10]

$$G : \forall x. P(x)$$

$$1 : (\exists z. R(z)) \rightarrow (\exists w. T(w))$$

$$2 : \forall x \exists y. \left((\forall z. (P(x) \vee R(z))) \wedge (\exists w. (T(w) \rightarrow Q(y))) \right)$$

$$3 : \forall w \exists y. \left((\forall x. (P(x) \rightarrow T(w))) \wedge (\forall x. (Q(y) \rightarrow P(x))) \right)$$

mgu(most general unifier) –the mgu g of $\{E_i\}$ has the property that if s is any unifier of $\{E_i\}$, yielding $\{E_i\}s$, then there exists a substitution s' such that $\{E_i\}s = \{E_i\}gs'$.

Example: $s=\{x|A, y|B\}$ unifies $P(x,f(y))$ with $P(x,f(B))$ but the mgu is $\{y|B\}$